



$$\begin{aligned}
 C^{[0,1]} \hat{N}^{[2]} &= |0\rangle\langle 0|^{[0]} \otimes \hat{I}^{[1,2]} + |1\rangle\langle 1|^{[0]} \otimes C^{[1]} \hat{N}^{[2]} \\
 &= \hat{I} \otimes \hat{I} \otimes \hat{I} + |1\rangle\langle 1| \otimes |1\rangle\langle 1| \otimes (\hat{\sigma}_x - \hat{I})
 \end{aligned}$$